

QUESTION:10 STACK USING ARRAY

main.c

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Run

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  #define MAX 5
4  int stack[MAX];
5  int top = -1;
6  void push(int value) {
7      if (top == MAX - 1) {
8          printf("Stack Overflow\n");
9      } else {
10         top++;
11         stack[top] = value;
12         printf("%d pushed into stack\n", value);
13     }
14 }
15 void pop() {
16     if (top == -1) {
17         printf("Stack Underflow\n");
18     } else {
19         printf("Deleted element: %d\n", stack[top]);
20         top--;
21     }
22 }
23 void display() {
24     if (top == -1) {
25         printf("Stack is Empty\n");
26     } else {
```

Output

```
--- STACK USING ARRAY ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 1
Enter value: 5
5 pushed into stack

--- STACK USING ARRAY ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 1
Enter value: 6
6 pushed into stack

--- STACK USING ARRAY ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 3
Stack elements are:
```

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```
27     printf("Stack elements are:\n");
28     for (int i = top; i >= 0; i--) {
29         printf("%d\n", stack[i]);
30     }
31 }
32 }
33 int main() {
34     int choice, value;
35     while (1) {
36         printf("\n--- STACK USING ARRAY ---\n");
37         printf("1. Push\n");
38         printf("2. Pop\n");
39         printf("3. Display\n");
40         printf("4. Exit\n");
41         printf("Enter your choice: ");
42         scanf("%d", &choice);
43         switch (choice) {
44             case 1:
45                 printf("Enter value: ");
46                 scanf("%d", &value);
47                 push(value);
48                 break;
49             case 2:
50                 pop();
51                 break;
52             case 3:
53                 display();
54                 break;
55             case 4:
56                 exit(0);
57             default:
58                 printf("Invalid Choice\n");
59         }
60     }
61     return 0;
62 }
```